

Project Proposal

Project Title

SmartFin: An Agentic AI Multi-Agent Co-Pilot for Personal Financial Management

Project Sponsor

N/A

Project Members

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Overview

Managing personal finances effectively is a persistent challenge for individuals across income levels. The daily tasks involved — tracking expenditure, planning budgets, monitoring for suspicious transactions, assessing overall financial health, and working towards savings goals — are fragmented across different tools and mental models. Most people rely on a combination of banking apps, spreadsheets, and manual effort, none of which provide integrated, context-aware guidance. As a result, overspending goes unnoticed until month-end, savings goals are set without a realistic path to achievement, and anomalous transactions may be identified too late.

SmartFin addresses this problem by providing an agentic AI co-pilot that unifies the key activities of personal financial management into a single, coordinated multi-agent system. The system is composed of five specialised agents — Expense Analysis, Budget Planning, Financial Goal Planning, Transaction Anomaly Detection, and Financial Health and Risk Assessment — orchestrated by a central coordinator that manages state, sequences agent activity, and surfaces outputs and alerts to the user through a unified interface. Each agent operates autonomously within its defined scope, draws on shared financial context maintained across sessions, and contributes to a coherent, continuously updated picture of the user's financial situation.

The business problem solved by this system is the absence of an integrated, intelligent layer that connects day-to-day financial behaviour with longer-term financial wellbeing. Existing personal finance tools are either passive record-keepers or narrow rule-based alert systems. SmartFin introduces agentic behaviour — proactive analysis, goal-oriented planning, anomaly reasoning, and health assessment — that goes beyond what any single tool or monolithic chatbot can provide.

Scope of Work

Agent Roles

The system comprises five specialised agents and one orchestrating agent, defined as follows:

- **Expense Analysis Agent:** Processes the user's transaction history, categorises expenditure across defined categories such as food, transport, housing, and entertainment, identifies spending trends over time, and flags instances where spending in a given category has deviated significantly from the user's established pattern. This agent serves as the primary data layer for the system and feeds its outputs to the Budget Planning Agent, the Financial Goal Planning Agent, and the Financial Health and Risk Assessment Agent.
- **Budget Planning Agent:** Operates at the level of the current period — typically a monthly cycle. Using the spending patterns surfaced by the Expense Analysis Agent, it generates or updates category-level budget allocations, monitors progress against those allocations within the period, and issues warnings when the user is on track to exceed a budget boundary. The Budget Planning Agent is concerned with short-term spending control: it answers the question of whether the user is living within their means in the immediate term and what adjustments are needed to do so.
- **Financial Goal Planning Agent:** Operates at a longer time horizon than the Budget Planning Agent and is concerned with the user's named financial objectives rather than period-by-period spending control. The user defines one or more goals — for example, accumulating a specific sum for a housing deposit, building an emergency fund, or saving for a planned purchase — along with a target amount and target date. The Financial Goal Planning Agent calculates the required monthly savings rate to achieve each goal, compares this against available surplus derived from the Budget Planning Agent's outputs, tracks cumulative progress, identifies conflicts between competing goals, and proactively notifies the user when their current trajectory will cause them to fall short. The distinction between this agent and the Budget Planning Agent is therefore one of scope and time horizon: the Budget Planning Agent governs how money is spent within the current period, while the Financial Goal Planning Agent governs whether money is being directed towards the user's future objectives across multiple periods.
- **Transaction Anomaly Detection Agent:** Monitors the user's transaction stream for patterns that deviate from the user's own established baseline. Anomalies flagged by this agent include transactions in unusual geographic locations, atypical transaction amounts relative to the user's history for a given merchant category, unusual transaction frequencies within a short time window, and transactions occurring at times inconsistent with the user's typical activity. This

agent does not function as an institutional fraud prevention system; it is positioned as a personal anomaly alerting layer that draws attention to transactions warranting the user's review. All flagged transactions are presented to the user with an explanation of the basis for the flag.

- **Financial Health and Risk Assessment Agent:** Provides a periodic assessment of the user's overall financial resilience and risk exposure, operating at a macro level above the period-by-period view of the Budget Planning Agent. It evaluates indicators including the ratio of total monthly obligations to income, the adequacy of liquid reserves relative to monthly expenditure, the presence of income concentration risk where salary constitutes the sole income source, and sustained trends of expenditure exceeding income. It produces a structured health summary with a qualitative rating and a set of prioritised observations, and it identifies systemic risks in the user's financial position that would not be visible from any single period's budget review.
- **Orchestrator Agent:** Coordinates the sequencing and state management of all five specialised agents using a LangGraph Supervisor architecture. It maintains a shared state object containing the user's transaction history, budget allocations, goal definitions, and prior agent outputs. It determines which agents to invoke in response to user actions or scheduled triggers, passes relevant context to each agent, aggregates outputs, and manages the delivery of notifications and summaries to the user. Human-in-the-loop checkpoints are enforced by the Orchestrator at moments where agent recommendations require user confirmation before being persisted, such as the initial setting of budget allocations or the registration of a new financial goal.

Effort Estimates

The following Work Breakdown Structure lists the major tasks and their estimated efforts. Estimates are in person-hours.

- **1. Project Setup and Architecture Design (14 hrs):** Define system architecture, select technology stack, establish repository structure, configure development environment, and document architectural decisions.
- **2. Orchestrator and Shared State Implementation (12 hrs):** Implement the LangGraph Supervisor Orchestrator, define the shared state schema, configure inter-agent context passing, implement Human-in-the-Loop checkpoint logic, and implement the notification output layer.
- **3. Expense Analysis Agent (12 hrs):** Implement transaction ingestion and normalisation, expenditure categorisation logic, trend analysis, and period-over-period deviation detection.

- **4. Budget Planning Agent (10 hrs):** Implement period-level budget allocation generation, in-period progress monitoring, and budget boundary warning logic drawing on Expense Analysis outputs.
- **5. Financial Goal Planning Agent (12 hrs):** Implement goal registration, required savings rate calculation, trajectory tracking, multi-goal conflict identification, and proactive shortfall notification logic.
- **6. Transaction Anomaly Detection Agent (12 hrs):** Implement baseline profiling from transaction history, anomaly detection rules covering geographic, amount, frequency, and timing dimensions, and explanation generation for flagged transactions.
- **7. Financial Health and Risk Assessment Agent (10 hrs):** Implement debt-to-income ratio calculation, liquidity reserve assessment, income concentration risk evaluation, sustained overspending trend detection, and health summary generation.
- **8. Guardrails and Security Implementation (8 hrs):** Implement input-layer prompt injection screening, execution-layer tool access enforcement, and output-layer schema validation across all agents.
- **9. Observability and LLMOps Integration (6 hrs):** Integrate LangSmith for execution tracing, configure monitoring, and implement prompt and model versioning configuration.
- **10. Testing (14 hrs):** Develop unit tests for all agent behaviours, integration tests for inter-agent state passing, end-to-end pipeline verification, and AI security tests.
- **11. UI Prototype (10 hrs):** Build a Streamlit-based web interface enabling transaction data input, display of agent outputs and alerts, goal registration, and Human-in-the-Loop confirmation interactions.
- **12. Documentation and Reporting (10 hrs):** Produce system architecture document, agent design documentation, responsible AI report, and AI security risk register.
- **13. Presentation Preparation and Demo (8 hrs):** Prepare presentation slides, script and rehearse demo scenarios, and finalise all project deliverables.

Total Estimated Effort: approximately 138 person-hours.